



K S Tharunpranav 251801359-2025-AIDS ▾

K2

Started on	Thursday, 20 August 2026, 8:33 AM
State	Finished
Completed on	Thursday, 20 August 2026, 8:39 AM
Time taken	5 mins 54 secs
Marks	1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int n, count = 0;
6      if (scanf("%d", &n) != 1) return 0;
7
8      int i = 1; count++;
9      int s = 1; count++;
10
11     while(s <= n)
12     {
13         count++;
14         i++; count++;
15         s += i; count++;
16     }
17     count++;
18
19     printf("%d", count);
20     return 0;
21 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)